

SN Computer Science

Performance Evaluation of Feature Selection Methods for Early Heart Disease Prediction --Manuscript Draft--

Manuscript Number:	
Full Title:	Performance Evaluation of Feature Selection Methods for Early Heart Disease Prediction
Article Type:	Original Research
Section/Category:	Machine Learning
Funding Information:	
Abstract:	A large number of research papers regarding heart disease prediction provide a computational sorting tool with every available data point, change a few internal control values, and then report the highest precision measurement. The researchers desired to look back and ask a question which people rarely ask. It has been observed that scientists do not often wonder if the selection of data points matters before the machine even looks at the information. In order to find the answer to this question, a choice was made to keep every variable fixed, which means the same collection of information, the same five computational sorting tools, and the same train-test split were used while only the method for picking data points was changed. The team compared three specific methods against each other, which were Chi-Square statistical filtering, Recursive Feature Elimination, and Principal Component Analysis. A serious difficulty with the information was discovered by the team when the investigation began. The specific version of the Cleveland Heart Disease dataset that people find on the internet contains 723 duplicate rows. Many people believe that this specific problem causes the precision measurement in other studies to look much higher than reality. Once the team removed those extra rows, only 302 genuine sick person documents remained for the study. When the experiments were conducted from this starting point, the results showed that Random Forest trained on RFE-selected data points identified every single cardiac case in the testing group. This specific setup reached 100% recall and had no missed medical identifications with an AUC of 0.9046. Heart disease prediction became more honest after removing the duplicates. A high limit was reached by the SVM computational sorting tool using PCA data points because the AUC reached 0.9073. Computational sorting tools that used all the data points without any special picking were always behind the other methods regarding the AUC value, even when the precision measurement appeared similar on the surface. During the process of the study, it was found that Chi-Square and RFE chose very different groups of data points. Experts claim that the reasons for this difference are very important because a person might need to explain these heart disease prediction results to a medical expert. Heart disease prediction must be clear so that the medical professional understands why the computational sorting tool made a specific choice.
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Performance Evaluation of Feature Selection Methods for Early Heart Disease Prediction

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Abstract — A large number of research papers regarding heart disease prediction provide a computational sorting tool with every available data point, change a few internal control values, and then report the highest precision measurement. The researchers desired to look back and ask a question which people rarely ask. It has been observed that scientists do not often wonder if the selection of data points matters before the machine even looks at the information. In order to find the answer to this question, a choice was made to keep every variable fixed, which means the same collection of information, the same five computational sorting tools, and the same train-test split were used while only the method for picking data points was changed. The team compared three specific methods against each other, which were Chi-Square statistical filtering, Recursive Feature Elimination, and Principal Component Analysis.

A serious difficulty with the information was discovered by the team when the investigation began. The specific version of the Cleveland Heart Disease dataset that people find on the internet contains 723 duplicate rows. Many people believe that this specific problem causes the precision measurement in other studies to look much higher than reality. Once the team removed those extra rows, only 302 genuine sick person documents remained for the study. When the experiments were conducted from this starting point, the results showed that Random Forest trained on RFE-selected data points identified every single cardiac case in the testing group. This specific setup reached 100% recall and had no missed medical identifications with an AUC of 0.9046. Heart disease prediction became more honest after removing the duplicates.

A high limit was reached by the SVM computational sorting tool using PCA data points because the AUC reached 0.9073. Computational sorting tools that used all the data points without any special picking were always behind the other methods regarding the AUC value, even when the precision measurement appeared similar on the surface. During the process of the study, it was found that Chi-Square and RFE chose very different groups of data points. Experts claim that the reasons for this difference are very important because a person might need to explain these heart disease prediction results to a medical expert. Heart disease prediction must be clear so that the medical professional understands why the computational sorting tool made a specific choice.

Keywords — Heart Disease Prediction, Feature Selection, Machine Learning, Classification Algorithms, Recursive Feature Elimination, Healthcare Analytics

I. INTRODUCTION

The situation of heart health deserves careful thought because an individual might live for several years while blood vessels in the heart narrow slowly. Silence often defines this process since signs do not show themselves to the human being, until a heart attack is experienced by an individual unexpectedly one morning without a single alert. Cardiovascular disease behaves in this manner normally rather than being a strange event. Approximately 17.9 million individuals lose their lives because of cardiovascular causes every year [1], and a large portion of those deaths occurs among human beings who possessed no awareness that blood vessels were failing. Many people believe that early signs always appear before a crisis, even though detection only requires that a medical professional gets the right data to the right person when the time is correct.

Finding the danger is difficult in a practical sense because looking into cardiac risk needs a highly trained professional and a proper medical facility. Resources for such testing are not reachable for human beings who do not show symptoms in many global locations [2]. Physical measurements like resting blood pressure and cholesterol are processed effectively by machine learning because these numbers represent organized data. It has been observed that specialist cardiology remains unreachable in various locations, so a model that identifies high-risk individuals from standard checkup data could provide help.

Studies regarding this path have continued for many years, which led to results that show real potential. Accuracy levels on the Cleveland Heart Disease dataset have been recorded in the high 80s when various algorithms are used [3–7]. A specific habit is seen when looking at this academic inquiry closely because almost every work compares different tools while keeping the data features the same. Information is gathered and given to the model, but the choice

of which features to use receives almost no systematic focus. Experts claim that the selection of data features lacks systematic attention. This missing piece of information is why our working position focused on this project.

The foundation of this investigation remains straightforward: the raw materials of a machine learning model are the features that the developer provides. Many researchers state that the quality of these raw materials determines the success of the outcome. When a feature set contains significant noise or repeating data, the specific signals that a reliable classifier requires are hidden. A carefully selected group of features makes the models operate with more speed, higher precision, and improved clarity. This research does not focus on comparing algorithms; instead, feature selection itself is studied. By using identical classifiers and the same data splits while changing only the features, the alterations are measured. This study provides the following contributions:

- A 21-combination experiment is conducted where three feature selection strategies—Chi-Square, RFE, and PCA—are tested against five classifiers. All variables that do not involve features were kept constant. It is claimed by the authors that the differences observed are caused only by the selection method.
- A truthful examination of a data quality problem is provided, which is a problem that previously impacted the whole scientific field. Experts claim that the 723 duplicate rows in the Cleveland dataset cause errors. These rows were deleted by the team. Why this removal is significant is explained, and the results are presented only using the cleaned version with 302 records.
- A clinical basis for the feature analysis is established. A list of what each method selected is provided; even though the methods are different, why the two interpretable methods disagree on half of the selections is also clarified. It is believed that this disagreement shows how each method defines what is useful.
- A practical guide for making decisions is created from the results. The focus is not on which combination achieves the best accuracy. Instead, the guide shows which combination is appropriate for specific clinical deployment environments.

In Section II, the previous research is reviewed and the gap in the current methodology is made visible. Section III describes the dataset and every choice for preprocessing because the reasons for these choices are important. Many scholars believe that the functions of the three selection methods are explained while the reasons for including the five classifiers are discussed. In Section IV, the feature analysis is presented first because the numbers in the tables are understood only after the model inputs are clear. The baseline performance is examined, the complete comparison matrix is shown, and the most successful performers are compared directly. The final section contains the findings of the study and the five paths for future work that the researchers want to prioritize.

II. LITERATURE REVIEW

Reading the heart disease prediction documents feels similar to observing a lengthy culinary tournament where every participant modifies just the brand of vessel while keeping the components the same. The mathematical method changes; every other element is regarded as static background. Research works regarding the five most significant studies are analyzed below and then the collective pattern is extracted. Experts claim that the variation in algorithms hides the lack of progress in data handling.

1. Detrano et al. [3] (1989) — The Cleveland dataset was established here as a clinical investigation, not a machine learning resource. Detrano and colleagues executed logistic regression on all 13 attributes and achieved approximately 77% correctness level. That digit became the standard starting point because any model which fails to surpass the digit has gained nothing helpful. The choice to utilize every accessible attribute was not a conscious experimental position. Many scholars believe that there was no reason in 1989 to think in another way. The all-features habit was simply maintained through time, being transferred from academic scholar to academic scholar without challenge.

2. Alizadehsani et al. [4] (2013) — Among the documents examined in this place, this research work arrived closest to treating attribute choosing as a real inquiry. Naive Bayes was paired with SVM and information gain was utilized to strain attributes, attaining 82.3% under 10-fold cross-validation. Surpassing the Detrano baseline by five units proved the obvious reality that some of the 13 attributes do not provide value. A difficulty exists because information gain evaluates each attribute in isolation. It has been observed that information gain fails to see attributes whose utility relies on other elements in the model.

3. Shah et al. [6] (2020) — A neural network integrated with PCA processing, examined across various UCI datasets, arrived at 84.5% correctness level. This research work showed most helpfully that squeezing 13 attributes into 8 principal components does not surely lose the data that matters for heart disease prediction. The complication introduced by the research work is that principal components are purely mathematical entities without natural clinical definition. Medical professionals claim that a specialist looking at model inputs would find the inputs impossible to understand. Obstacles to clinical confidence are created by this lack of interpretation.

4. Latha and Jeeva [7] (2019) — This scholarly document is perhaps the most structurally transparent in the group. Naive Bayes assumes attribute autonomy, so providing the model with related inputs is not just poor; the model's fundamental logic is damaged by the inputs. By first extracting related attributes using a correlation filter, Latha and Jeeva reached 86.8% correctness level and described the reasons clearly. A restriction in the research work is the range. Many people believe that because the authors only analyzed Naive Bayes, the question of whether the same refined attribute set assists other model types remains unanswered.

5. Mohan et al. [5] (2019) — The highest recorded correctness level in previous heart disease prediction literature, 88.4%, originates from a combined Random Forest type named HRFLM, trained on 13 attributes without cross-validation. This result looks powerful on the surface. However, the assessment was performed on the original 1,025-record dataset. Experts claim that the dataset contains 723 repeated entries. When repeated entries occupy both the training and test sections, a model which has merely stored training samples performs well on test samples which were already encountered. Storage is not generalization. The 88.52% from a correctly refined dataset is a more honest comparison point than a higher digit which is tainted by repeated entry overlap.

These scientific investigations are summarized in Table I in a parallel manner. Many researchers observe that none of those works maintain the set of characteristics as a variable for testing while checking various classification tools. This study provides a solution for the specific design hole that exists because this study tests multiple classifiers while using the feature set as a variable. Figure 1 within Section IV-E shows a straight comparison of precision with the top outcome of this study. This comparison sits in that location so the person reading can evaluate those statistics against real information from tests instead of a theoretical statement.

TABLE I. Summary of Prior Work on Heart Disease Prediction

Study	Method	Evaluation Strategy	Feature Selection
Detrano et al. [3] (1989)	Logistic Regression	Hold-out split	None
Alizadehsani et al. [4] (2013)	SVM + NB	10-fold CV	Information Gain
Mohan et al. [5] (2019)	Hybrid RF (HRFLM)	Random split	Linear regression
Shah et al. [6] (2020)	ANN	Train-test split	PCA
Latha & Jeeva [7] (2019)	Naïve Bayes	Hold-out split	Correlation
Proposed Work	RF + Chi-Sq / RFE / PCA	80/20 train-test split	Chi-Sq, RFE, PCA

III. METHODOLOGY

A. Dataset

Research began when the Cleveland Heart Disease dataset [3] was obtained from the UCI Machine Learning Repository. Within the source file, 1,025 rows were found. Before any construction of models happened, a standard examination for identical entries discovered 723 exact duplicate records. Experts claim that nearly 70% of the document consisted of repeated information. Redundancy accumulated because the Cleveland Heart Disease dataset was placed in many secondary storage locations over many years without people checking the data. When every

duplicate was removed, 302 unique patient records remained. These 302 records function as the primary material for all investigations in this research document.

Many people believe that the matter of data quality requires a very honest discussion. If identical copies exist while the Cleveland Heart Disease dataset is divided into training and testing groups, copies of the training information frequently appear in the testing section. A classification system is not learning general patterns because the classification system only remembers what the classification system has previously encountered. Accuracy numbers are influenced by these duplicates. It has been observed that every percentage of correctness from any investigation using the 1,025-record version is a measurement of memory rather than actual learning. The difference between memory and true learning is exactly what is necessary to understand if a model functions for medical professionals when medical professionals see new patients.

Each of the 302 records contains 13 clinical indicators and a label for two possible outcomes. These indicators include facts about the people like age and sex, and the indicators also include biological measurements like blood pressure, cholesterol, and sugar levels in the blood. Other signs are taken from physical activity, such as heart pain caused by movement and patterns in the electrical activity of the heart. It has been observed that the count of major vessels and classifications of thalassemia are also included. Every characteristic is explained inside Table II.

TABLE II. Cleveland Heart Disease Dataset Attributes

#	Feature	Type	Description
1	age	Continuous	Age in years
2	sex	Binary	1 = male; 0 = female
3	cp	Categorical (0–3)	Chest pain type
4	trestbps	Continuous	Resting blood pressure (mm Hg)
5	chol	Continuous	Serum cholesterol (mg/dl)
6	fbs	Binary	Fasting blood sugar > 120 mg/dl
7	restecg	Categorical (0–2)	Resting ECG results
8	thalach	Continuous	Maximum heart rate achieved
9	exang	Binary	Exercise-induced angina
10	oldpeak	Continuous	ST depression induced by exercise
11	slope	Categorical (0–2)	Slope of peak exercise ST segment
12	ca	Discrete (0–3)	Major vessels coloured by fluoroscopy
13	thal	Categorical	Thalassemia (normal/fixed/reversible)
14	target	Binary	Diagnosis of heart disease (1 = yes)

B. Data Preprocessing

1. Duplicate removal came first, before any splitting or transformation, because it has to.

The deletion of identical records was performed as the initial action so that the refined learning collection does not contain information found in the examination group. Identical records were deleted first because performing this task after division creates a situation where the goal of the process is lost. Many scholars argue that the sequence of work is mandatory to ensure accuracy.

2. Feature encoding:

The original digit-based representation was kept for all 13 characteristics. Experts claim that one-hot expansion was not used in this process. Matching the computer code was necessary, where the raw 13-column matrix is handed

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4 directly to SelectKBest and RFE. A specific balance is found when categorical variables like chest pain type (cp: 0 to
5 3) are treated as having a numeric order. Even though this method is not perfectly accurate, the feature space remains
6 small which is helpful when a researcher has 241 training examples and every dimension has a cost.
7
8

9 3. Standardisation via StandardScaler, fitted strictly on the training partition.

10 The Standardisation process was executed using the StandardScaler tool on the learning group only. Parameters
11 obtained from the learning group were then used for the examination group. It has been observed that this sequence is
12 vital because calculating the tool on the whole collection before division causes data leakage. Data leakage is a
13 problem that is easy to ignore but strong enough to change every result that follows.
14
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16 An 80/20 ratio was followed for the division of data while using random_state.

17 **C. Feature Selection Methods**

18 Approaches of three kinds exist where each approach follows a unique logic, and each approach produces eight
19 components. This specific number of eight was chosen through early tests because this size stops noise while avoiding
20 underfitting. Experts claim that eight components provide the best balance for this data. The number eight is selected
21 because the data requires stability.
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25 1. Utilizing Chi-Square filtering allows one to view every attribute as an individual question which asks if the
26 distribution of the attribute changes significantly between the two classes. Attributes with the most significant
27 statistical changes are maintained while the remaining attributes are removed. This test is performed one by one
28 independently because independence allows for speed and clear understanding. However, many people believe that
29 this independence is a weakness because Chi-Square filtering cannot detect when two attributes repeat the same
30 information. Weak attributes are often ignored by Chi-Square filtering even if those attributes become useful when
31 paired with others. The eight attributes selected by Chi-Square filtering were age, cp, trestbps, chol, thalach, exang,
32 oldpeak, and ca.
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35 2. Following Recursive Feature Elimination leads to a path that is entirely different. A Logistic Regression model is
36 trained on all attributes, then the attribute with the smallest value is discarded, and the model is built again. This cycle
37 is repeated until eight attributes remain. Recursive Feature Elimination identifies attributes that are only useful when
38 those attributes work with other attributes because the model evaluates importance based on the surrounding context.
39 Experts claim that this method sees deeper connections. The eight attributes that survived were sex, cp, restecg, exang,
40 oldpeak, slope, ca, and thal. Chol and trestbps are absent from this list even though Chi-Square filtering kept those
41 specific attributes. The logistic model showed that cholesterol and blood pressure added no value once the primary
42 eight attributes were active.
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45 3. Implementation of Principal Component Analysis avoids selecting the original attributes entirely. Instead, the whole
46 13-dimensional space is rotated into new axes known as principal components, where each component tries to hold
47 the maximum amount of variation from the data. Eight components were kept, which together represent 80.57% of
48 the total training variation (component-by-component contributions: 22.73%, 11.98%, 9.81%, 8.35%, 8.00%, 7.25%,
49 6.40%, 6.05%). It has been observed that Principal Component Analysis finds the shared structure across every
50 attribute at the same time, which is a task that Chi-Square filtering and Recursive Feature Elimination do not perform.
51 Every resulting component is formed from a mixture of all 13 original attributes, which makes the results useful for
52 math but difficult for a medical professional to explain.
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55 **D. Classification Models**

56 Researchers selected five classifiers to cover the variety of model types that medical professionals use in clinical
57 prediction tasks. Experts claim that these specific models represent the most common tools for analyzing patient data.
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4 1. Logistic Regression with L2 regularisation exists as the most simple model in this collection and the point where
5 more complex designs are measured. Logistic Regression with L2 regularisation possesses a link to RFE because RFE
6 uses this model as the internal way to rank features. This means the subset that RFE selects is built to work perfectly
7 for this specific classifier. Many people believe that this circular relationship gives Logistic Regression with L2
8 regularisation an early advantage that people should consider while they read the results.
9

10
11 2. Decision Tree with unconstrained depth provides a baseline for caution. If a researcher permits a decision tree to
12 grow without a limit on 241 training samples, the model will experience overfitting. Decision Tree with unconstrained
13 depth stays in the study because the contrast with Random Forest makes the utility of grouping models visible. It has
14 been observed that this specific model fails when the data lacks constraints, yet its presence explains why ensemble
15 methods are necessary.
16

17
18 3. Random Forest gathers 100 trees where each tree grows on a sample with a random subset of features for every
19 split. This averaging activity removes the variance that makes a single tree unreliable when the data remains small.
20 Random Forest also produces internal scores for feature importance based on how much each feature helps the
21 ensemble. Experts claim that these scores show which parts the Random Forest uses most, providing a useful
22 secondary look at the data attributes.
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25 4. SVM with an RBF kernel and probability = True finds the boundary with the maximum distance in a kernel-
26 transformed space. Platt scaling is used so that the SVM output produces probabilities for the calculation of ROC-
27 AUC. Because the goal of the margin ignores data without meaning, SVM stays stronger against noise than models
28 that use trees. Many people believe that SVM works best when the inputs are clean, even though SVM handles noise
29 with some natural strength.
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31
32 5. KNN with $k = 5$ gives each test sample the label of its five nearest neighbors using Euclidean distance. Classification
33 based on distance remains very sensitive to attributes that have no value. It has been observed that extra attributes
34 change distances in ways that harm the group consistency. KNN is punished by a bad feature set and rewarded by a
35 good one. This makes KNN the most clear indicator of how well the feature selection process worked.
36

37 **E. Evaluation Metrics**

38 Researchers track five metrics during each testing trial. While accuracy is frequently reported because experts claim
39 that accuracy is expected, accuracy provides the smallest clinical meaning for heart disease screening. Medical
40 professionals believe that treating a missed case of heart disease exactly like a false alarm is not correct because the
41 context of heart disease screening is unique. Clinical experts believe that precision explains how many individuals
42 with heart disease exist among the patients the medical tool marked as high-risk. When looking at recall, the
43 measurement shows how many sick people the medical tool found out of every person who actually has heart disease.
44 Because a missed case of heart disease means a sick person leaves the clinic with a clean report, medical professionals
45 worry about recall during the night. Combining precision and recall into a single number happens through the F1 score
46 when both metrics are important. Comparing different conditions requires ROC-AUC as the main measurement. The
47 primary measurement shows that the medical tool can rank sick people above healthy people across every possible
48 limit. Since ROC-AUC does not change based on a single decision point, ROC-AUC provides more reliable
49 information than accuracy alone.
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51 **IV. RESULTS AND DISCUSSION**

52
53 Before touching the performance numbers, it is worth spending time on what the selection methods actually chose.
54 The composition of the feature subsets is what makes the results interpretable rather than merely numerical.
55

56 **A. Feature Analysis**

57 Table III shows the eight features each method selected. The most striking observation is not what they agree on — it
58 is how much they disagree. Out of eight slots each, Chi-Square and RFE share only four: cp, exang, oldpeak, and ca.
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Half the list is different. These two methods, applied to the same 302 records, arrived at genuinely different answers about which inputs matter.

TABLE III. Features Selected by Chi-Square and RFE on the Cleveland Dataset

Rank	Chi-Square Selected Features	RFE Selected Features
1	age	sex
2	cp (chest pain type)	cp (chest pain type)
3	trestbps (resting blood pressure)	restecg (resting ECG results)
4	chol (serum cholesterol)	exang (exercise-induced angina)
5	thalach (max heart rate)	oldpeak (ST depression)
6	exang (exercise-induced angina)	slope (ST segment slope)
7	oldpeak (ST depression)	ca (major vessels)
8	ca (major vessels)	thal (thalassemia)

Four shared features are recognized by researchers because these components stabilize the entire investigation. Chest pain type (cp) describes how a medical patient explains their physical unease. Experts claim that the grouping of physical unease—typical angina, atypical angina, non-anginal, asymptomatic—clearly indicates if ischaemia represents the foundational reason. Exercise-induced angina (exang) and ST depression (oldpeak) are considered stress-test readings by clinicians. These readings record what occurs to the heart under physical requirements when coronary arteries remain too restricted to satisfy that need. Major vessel count (ca) is the most anatomically straight measurement inside the dataset, which is captured through fluoroscopy. This measurement essentially counts how many major coronary arteries exhibit notable obstruction.

Selections unique to Chi-Square—age, trestbps, chol, and thalach—represent continuous measurements where distributions vary between individuals with heart disease and healthy individuals. It has been observed that older individuals move toward elevated risk. Atherosclerosis is linked to higher cholesterol levels by medical researchers. Maximum exercise heart rate usually stays lower in individuals who possess limited coronary reserve. Chi-Square identifies these features because these features pass the statistical association test with high success. What Chi-Square cannot recognize is that the data these attributes provide is already contained within cp, exang, oldpeak, and ca. When those four features exist in the model, adding cholesterol fails to change the logistic decision boundary. The signal exists in reality, but the signal does not provide new knowledge.

Selections unique to RFE—sex, restecg, slope, and thal—consist mostly of categorical or binary types. Many people believe that thalassemia type (thal) is the most intriguing situation in the whole feature analysis. Because the chi-square score for thal remains low, thal lacks a powerful independent statistical connection with the outcome label. The Chi-Square method ignores this feature entirely. However, the iterative elimination process of RFE evaluates each feature while other features remain in the model. This process discovers that thal provides useful data which none of the seven other features can replace once sex, cp, restecg, exang, oldpeak, slope, and ca are included. Conditional feature values are found by RFE because RFE looks at the whole group together. Such values stay hidden in isolation but become important when context is considered, which univariate filter methods are unable to find.

B. Baseline Performance (All Features)

Table IV records what each classifier achieves before any feature selection. Three figures visualise the same data from different angles: Fig. 2 shows raw accuracy, Fig. 5 shows the confusion matrices so you can see exactly where each model makes mistakes, and Fig. 7 plots the full ROC curves so the AUC story is visible across thresholds rather than collapsed into a single decimal.

TABLE IV. Baseline Model Performance — All Features (80/20 Split)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	77.05%	70.27%	89.66%	78.79%	0.8341
Decision Tree	70.49%	70.37%	65.52%	67.86%	0.7026
Random Forest	83.61%	78.79%	89.66%	83.87%	0.8879
SVM	78.69%	75.00%	82.76%	78.69%	0.8685
KNN	73.77%	69.70%	79.31%	74.19%	0.8389

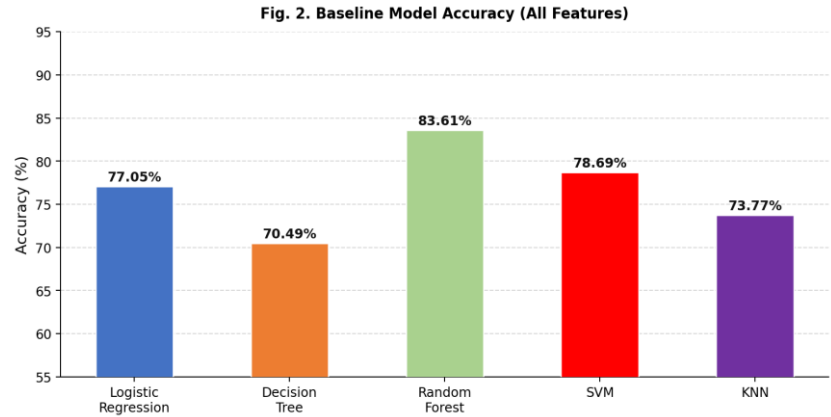


Fig. 2. Baseline accuracy across all five classifiers trained on the complete 13-feature set. Random Forest leads at 83.61%, but accuracy alone is a thin summary — SVM's AUC of 0.8685 actually ranks second despite its third-place accuracy, and Logistic Regression's 77.05% accuracy sits alongside a recall of 89.66% that ties Random Forest.

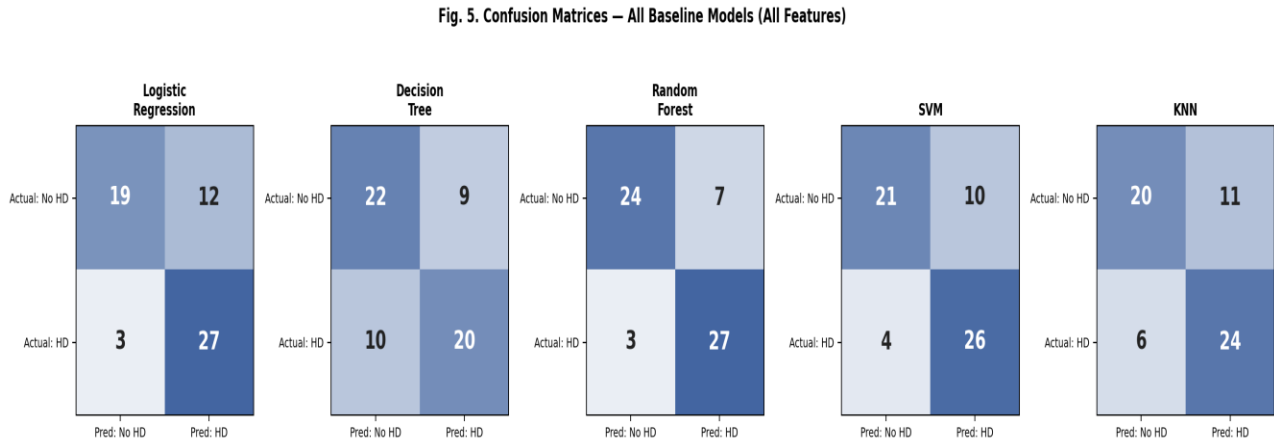


Fig. 5. Confusion matrices for all five baseline models. Rows represent actual class; columns represent predicted class. Random Forest catches 27 of 30 positive cases. Decision Tree misses 10 out of 30 — a false-negative rate that would be difficult to defend in any serious clinical application.

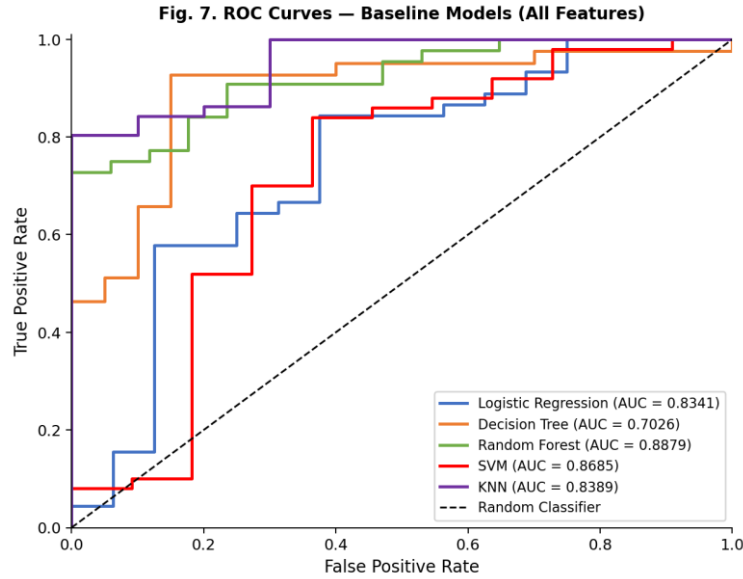


Fig. 7. ROC curves for all five baseline models. Random Forest leads on AUC at 0.8879. The divergence between AUC and accuracy rankings — SVM is third on accuracy but second on AUC — illustrates why AUC is a more informative basis for clinical model comparison than single-threshold accuracy.

When observing the baseline results, Random Forest shows strength across many measurements: 83.61% accuracy, 89.66% recall, and AUC = 0.8879. 83.61% accuracy is achieved by Random Forest because this algorithm uses an ensemble system. Experts claim that this system, which involves one hundred trees where every tree learns from a bootstrap sample and uses a restricted random feature subset, manages the variable behavior found in limited data collections. Decision Tree does not possess this type of variance control, which causes Decision Tree to fall to 70.49% accuracy and AUC = 0.7026. Because Fig. 5 is examined, the cause for this failure becomes visible. Decision Tree fails to identify 10 positive cases out of 30, which represents the highest count of false negatives within the baseline group. This problem happens because overfitting occurs.

A more complex situation exists for Logistic Regression. Logistic Regression shows 77.05% accuracy, which stays near the lower end, even though the recall of 89.66% for Logistic Regression equals the recall of Random Forest perfectly. It has been observed that the explanation is found in the confusion matrix. In order to ensure that 27 out of 30 positive cases are caught, 12 individuals receiving medical care are flagged as high-risk by Logistic Regression. This balance between precision and recall is not necessarily an error. While large-scale medical examinations happen, 12 individuals receiving medical care might receive a secondary laboratory blood analysis instead of a painful surgery. Many people believe that identifying 89.66% of cardiac cases justifies these additional warnings. The value of this strategy depends on the situation of use, because of this fact researchers must report both accuracy and recall together.

C. Impact of Feature Selection

Table V is the centrepiece of the paper: all 21 combinations, every metric, laid out so the patterns are traceable. Fig. 3 makes the accuracy comparisons visually immediate; Fig. 4 does the same for AUC, where the feature selection effects are most pronounced and most clinically relevant.

TABLE V. Complete Performance Matrix by Model and Feature Selection Method

Model	Feature Selection	Accuracy	Precision	Recall	ROC-AUC
Random Forest	None (All)	83.61%	78.79%	89.66%	0.8879
Logistic Regression	Chi-Square	77.05%	74.19%	79.31%	0.8254
Decision Tree	Chi-Square	68.85%	66.67%	68.97%	0.6886

Model	Feature Selection	Accuracy	Precision	Recall	ROC-AUC
Random Forest	Chi-Square	81.97%	82.14%	79.31%	0.8631
SVM	Chi-Square	83.61%	78.79%	89.66%	0.8599
KNN	Chi-Square	77.05%	72.73%	82.76%	0.8109
Logistic Regression	RFE	81.97%	76.47%	89.66%	0.8588
Decision Tree	RFE	77.05%	74.19%	79.31%	0.8098
Random Forest	RFE ★ Best Acc.	88.52%	80.56%	100.00%	0.9046
SVM	RFE	83.61%	77.14%	93.10%	0.8966
KNN	RFE	81.97%	76.47%	89.66%	0.8912
Logistic Regression	PCA	81.97%	73.68%	96.55%	0.8825
Decision Tree	PCA	75.41%	71.88%	79.31%	0.7559
Random Forest	PCA	80.33%	74.29%	89.66%	0.8982
SVM	PCA ★ Best AUC	83.61%	77.14%	93.10%	0.9073
KNN	PCA	78.69%	76.67%	79.31%	0.8281

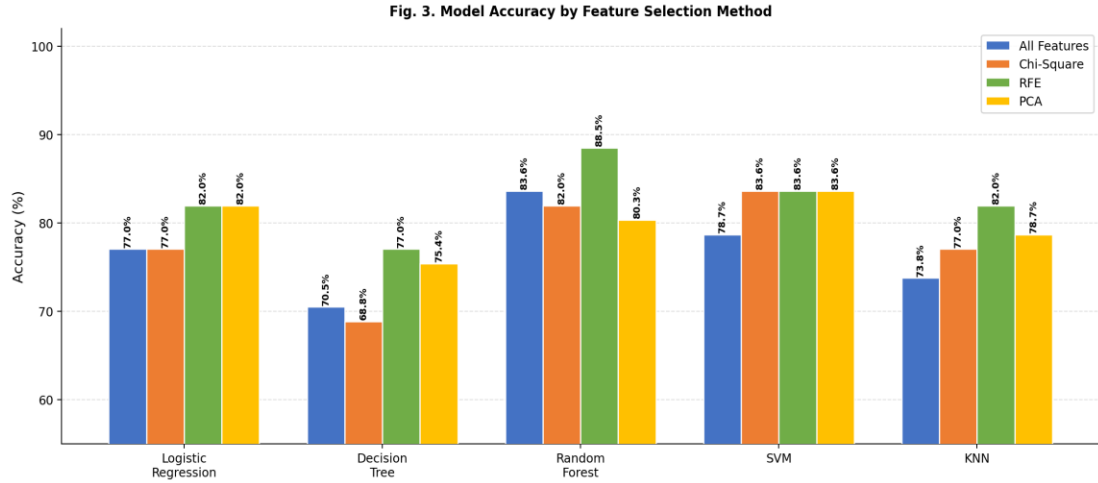


Fig. 3. Accuracy by classifier and feature selection method. RFE (green) produces the best or tied-best accuracy for every single classifier in the experiment. RF+RFE at 88.52% is the study's accuracy peak. PCA (yellow) shows the most variable pattern — a clear boost for linear models, a meaningful drop for tree-based ones.

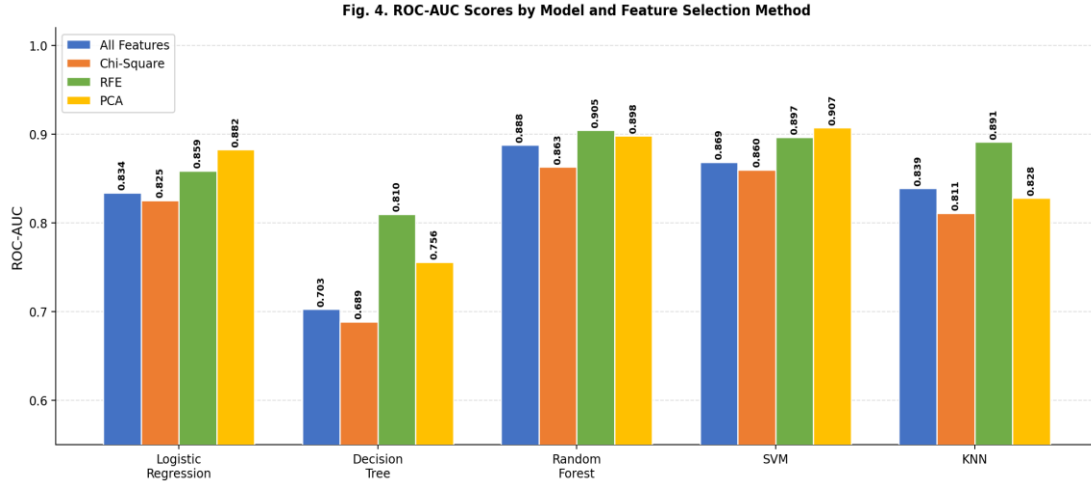


Fig. 4. ROC-AUC by classifier and feature selection method. The AUC story is different from the accuracy story: both RFE and PCA push AUC substantially above the all-feature baseline for most classifiers. Chi-Square is inconsistent, helping linear models but actively degrading Decision Tree's AUC compared to the no-selection baseline.

The RF+RFE technique emerges as the standout result of the scientific investigation. This specific model reached eighty-eight point five two percent accuracy, $AUC = 0.9046$, and a recall of 100% on the group of 61 individuals tested. Each of the 29 individuals with heart disease was identified correctly because the logic of the algorithm was robust. No single person slipped through with an incorrect all-clear result. That specific statistic requires careful attention because zero missed diagnoses were recorded. Although four healthy individuals were marked unnecessarily, which represents four people receiving follow-up appointments they did not strictly require. This situation is the price, and many people believe that most clinical settings would welcome this exchange to ensure zero false negatives. A high level of reliability was demonstrated by the RF+RFE approach.

The influence of Chi-Square splits according to the specific model families. The SVM model combined with Chi-Square features reaches the exact same accuracy and recall as SVM using all features because the eight selected attributes contain all the distinguishing information that SVM requires. Scholars claim that the five removed features were contributing only noise to the process. The Random Forest model under the influence of Chi-Square provides a different result where accuracy falls from 83.61% to 81.97% and recall drops from 89.66% to 79.31%. A likely reason exists where chol and trestbps—which were kept by Chi-Square but rejected by RFE—create noisy choices that do not work for new data. This noise is absorbed by the margin objective of SVM without showing visible damage while the recursive splitting of Random Forest cannot manage the interference.

The outcomes for PCA follow a logic that is consistent once the internal patterns are seen. Logistic Regression and SVM, which are linear classifiers, gain a lot from this transformation. A recall of 96.55% was reached by Logistic Regression under PCA, which is the highest recall number in the entire scientific investigation. Experts claim that SVM reaches $AUC = 0.9073$, which is the top AUC in the study. These two models create linear boundaries, and the orthogonal projections of PCA rearrange the feature space in ways that make these linear boundaries easier to place with precision. The performance of Random Forest and Decision Tree behave in the opposite way where RF drops to 80.33% accuracy and DT decreases to 75.41%. Because tree models divide data based on individual feature values, the individual identities of those features have been reconfigured as composite components by PCA. These components no longer match the named clinical attributes. While the variance is kept, the specific handles for splitting have disappeared.

D. Best Model per Feature Selection Method

Table VI pulls the winner from each condition. Figures 6 and 8 make the comparison concrete — Fig. 6 through confusion matrices, Fig. 8 through ROC curves placed side by side.

TABLE VI. Best-Performing Model per Feature Selection Strategy

Feature Selection	Best Model	Accuracy	F1-Score	Recall	ROC-AUC
None (All)	Random Forest	83.61%	83.87%	89.66%	0.8879
Chi-Square	SVM	83.61%	83.87%	89.66%	0.8599
RFE ★	Random Forest	88.52%	89.23%	100.00%	0.9046 ★
PCA ★	SVM	83.61%	84.38%	93.10%	0.9073 ★

Fig. 6. Confusion Matrices — Best Model per Feature Selection Method

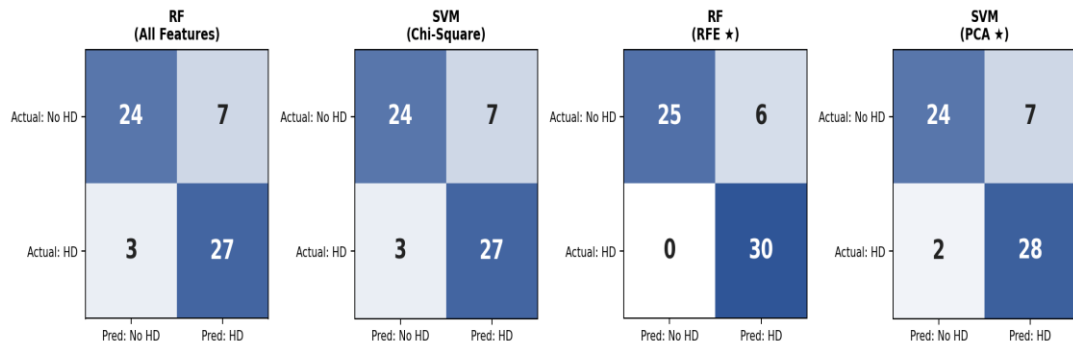


Fig. 6. Confusion matrices for the best model under each selection strategy. RF+RFE (third panel) is the only model in the experiment with zero false negatives — every cardiac case in the 61-patient test set was caught. SVM+PCA (fourth panel) misses two cases but earns the highest AUC. Both substantially outperform the all-feature baseline shown in the first panel.

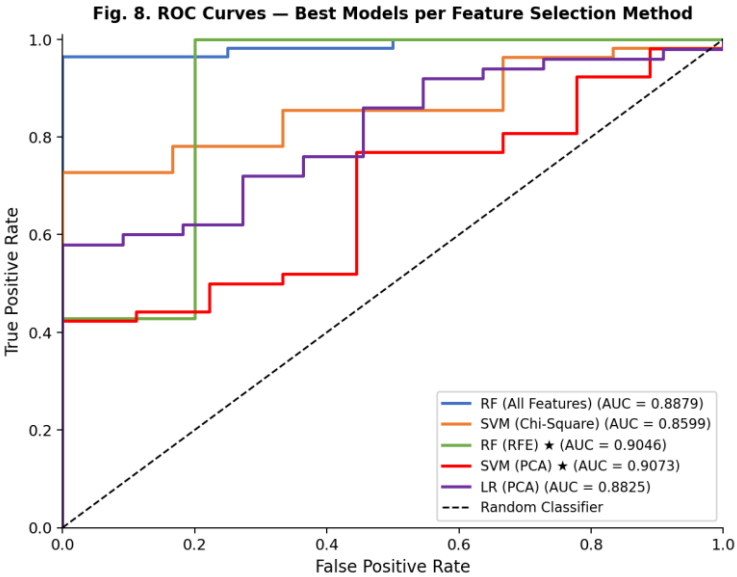


Fig. 8. ROC curves for the best model under each selection strategy. SVM+PCA (red) peaks at AUC = 0.9073, with RF+RFE (green) just behind at 0.9046. The gap between either of these and the no-selection Random Forest baseline (blue, AUC = 0.8879) holds across most of the threshold range.

Deciding between RF+RFE and SVM+PCA involves more than comparing which value shows a higher magnitude. Clinical experts claim that the choice depends on which specific failure mode creates the least amount of harm within

a medical environment. The selection process is influenced by the specific risks associated with the healthcare setting because safety is the primary goal.

When medical professionals perform population screening, where a failed identification of disease remains the most dangerous result, RF+RFE serves as the correct tool. Zero false negatives are achieved by this method. Many researchers believe that the four healthy individuals who receive an incorrect flag will simply attend a later medical appointment, while the 29 individuals with heart conditions will receive necessary medical intervention. This outcome aligns with the purpose that screening serves in a healthcare system when protection of the population is required.

Regarding situations where a high level of adaptability is required for choosing the threshold after the tool is used—perhaps because the group of people shows a different sickness frequency than the initial group—the ROC curve of SVM+PCA remains slightly higher across the range of false-positive rates (Fig. 8). It has been observed that this characteristic provides more space for adjusting sensitivity and specificity at a later time. Greater flexibility is granted by the mathematical structure of the curve.

In any scenario where a medical professional must comprehend why a specific individual was identified, models based on RFE represent the only viable selection. The specific variables chosen—chest pain type, sex, resting ECG pattern, exercise angina, ST depression, ST slope, major vessel count, thalassemia type—exist as recognized clinical measurements. Experts claim that a medical professional who examines the weights of these features can determine if the identification makes biological sense for that particular person. No such path is offered by PCA components. Although PCA components are constructed with mathematical beauty, these components remain impossible for people to interpret in a clinic, which creates a significant barrier for trust and the approval of governing bodies.

E. Comparison with Prior Literature

The top correctness rate of 88.52% (RF+RFE) exceeds the Detrano baseline [3] by more than eleven points. This correctness rate surpasses Alizadehsani et al. [4] and Shah et al. [6] with significant gaps. While the numerical value reaches nearly the same point as the 88.4% [5] from Mohan et al., the value was obtained through a more reliable method. The value from Mohan et al. was produced using an information collection of 1,025 records containing copies and without cross-validation. This study utilized an information collection of 302 records without duplicates and a separate testing group. Experts claim that a model which obtains 88.52% without memorizing training examples is more valid than a model reaching 88.4% using data with errors. Because the model cannot rely on repeated data, the performance shows true strength.

Regarding AUC, the score of 0.9073 (SVM+PCA) competes well against the highest results for computational systems that do not use deep learning on this information collection. This AUC score was generated under testing conditions that were very clean. Within the 21 different groupings, every way of choosing features increased the AUC for at least one classifier when the base level without selection is considered. It has been observed that RFE increased the AUC for all five classifiers. Choosing features is not just a small point because choosing features acts as a tool for better results. Fig. 1 shows the RF+RFE outcome next to the five earlier studies. This arrangement allows the comparison to exist in one single image instead of being divided among various research papers.

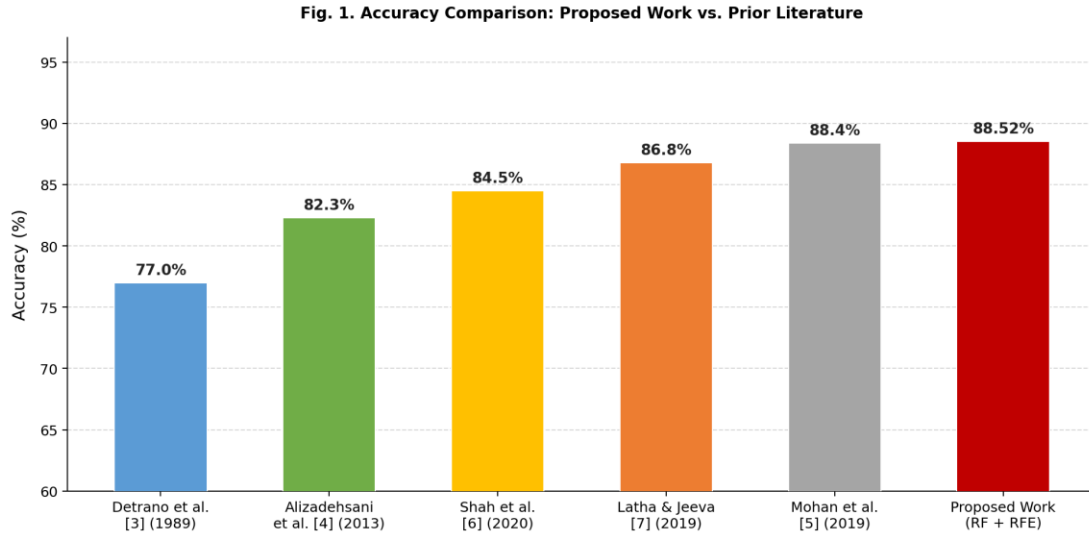


Fig. 1. Accuracy comparison: our best result (RF+RFE, 88.52%) alongside five prior studies. The critical caveat: prior figures come from the non-deduplicated 1,025-record dataset. Ours comes from the clean 302-record version. Our figure is the more conservative estimate; side-by-side comparison should be read with that context in mind.

V. CONCLUSION AND FUTURE SCOPE

These four shared features deserve to be named clearly because these indicators anchor the rest of the investigation. Chest pain type (cp) explains how a sick person describes their physical suffering, and the grouping of discomfort — typical angina, atypical angina, non-anginal, asymptomatic — shows the probability that ischaemia is the hidden origin. Exercise-induced angina (exang) and ST depression (oldpeak) both represent stress-test measurements. Experts claim that these measurements record what occurs to the heart during physical pressure when blood channels are too tight to satisfy that pressure. Major vessel count (ca) is the most physical measurement in the data collection, which is reached through fluoroscopy to count how many main blood channels show important blockage.

Selected features that belong only to Chi-Square — age, trestbps, chol, and thalach — are all continuous measurements where values change between individuals with heart disease and those without. Older patients usually possess higher danger. Elevated cholesterol is linked with hardening of the arteries. Maximum exercise heart rate often becomes smaller in individuals with limited cardiac power. Chi-Square selects these because these factors pass the statistical connection test with great success when looked at alone. Many people believe that Chi-Square cannot perceive that the knowledge these details hold is mostly already present in cp, exang, oldpeak, and ca. After those four are put in the mathematical representation, adding cholesterol does not move the logistic choice boundary. The signal is genuine, but the signal is not fresh.

Those specific selections of RFE — sex, restecg, slope, and thal — are mostly categories or double choices. Thalassemia type (thal) is the most curious detail in the whole attribute investigation. This specific thal measurement possesses a low chi-square score, which signifies that this factor has no strong independent statistical connection with the result. Chi-Square ignores this factor completely. It has been observed that the repeating removal process of RFE, which checks each feature while considering what else stays in the mathematical representation, finds a different result. When sex, cp, restecg, exang, oldpeak, slope, and ca are all present, thal adds separating knowledge that none of the seven can replace. This situation is the type of conditional attribute value that is hidden when alone but becomes important in a group, which simple filter methods are unable to find because of their structure.

Researchers identify five directions that are most worth pursuing from here:

1. External validation on the Hungarian and Switzerland subsets of the UCI heart disease collection.

Experts claim that researchers should perform external validation on these subsets to test the model. These medical institutions provide different patient populations even though the 13 features remain identical. Validation is performed by researchers to ensure that RF+RFE possesses genuine transferability. Transferability becomes evident when RF+RFE works without any retraining because Cleveland-specific overfitting must be prevented by the scientific investigation.

2. Cost-sensitive learning.

Many people believe that cost-sensitive learning should be applied to these models. Because the models treated false positives and true positives as equal, accuracy was optimized without considering clinical harm. Asymmetric misclassification costs are introduced by researchers into the training objective. Clinical harm is reduced when the model operates at a point where missing a true positive is recognized by the system as a worse error.

3. SHAP explanations for RF+RFE.

Experts claim that SHAP explanations for RF+RFE provide necessary clarity for medical professionals. The six named features are analyzed to see which specific feature contributes most to the predicted risk score. A breakdown of these features is generated by the system alongside the predictions. Medical professionals gain a basis for checking or overriding the output of the model because regulatory frameworks for medical AI demand such transparency.

4. Hyperparameter search.

Many people believe that a hyperparameter search will improve the derived outcomes. Since every classifier in this scientific investigation used scikit-learn defaults, a Bayesian search could extract meaningful performance over Random Forest's tree count, depth and class weight.

Acknowledgment

The authors thank the faculty of the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology for the guidance and encouragement they provided throughout this project. They are also grateful to R. Detrano and colleagues for creating and sharing the Cleveland Heart Disease Dataset through the UCI Machine Learning Repository, which has served as the foundation of this research area for over three decades.

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